2025 Invitational Tournament - Div C -Georgia

Disease Detectives

Written by: Abigail Bias, Courtney Coco, Ellin Park, Ally Xu

School Name/Team: ____

Student Names: ____

Instructions:

1. Write all answers on the Answer Sheet. DO NOT WRITE on Test Packet.
2. See the Image Sheet when asked to refer to an image by a question (it may be a separate sheet or attached to the back of this packet).
3. For numerical questions, answers within a reasonable range will be accepted.
4. When answering free-response questions, use the appropriate abbreviations for satellite and instrument names
5. For free-response questions, the answer sheet indicates the amount of space you should allocate to fully answer each question. Points are assigned based on accurately answering each part of the question.
6. Ties marked with a * will be broken in order

FOR GRADERS ONLY: ____________________ / 105 (Total points)

<u>Section 1: Definitions</u> *(16 points– 1 point per question)*

A. Descriptive epidemiology	B. Seasonality	C. Prevalence	D. Incubation Period
E. Ordinal Scale	F. Latency Period	G. Exposed Group	H. Cluster
I. Contagious	J. Frequency	K. Carrier	L. Necessary Cause
M. Sporadic Illness	N. Attack Rate	O. Epidemic	P. Infectivity

1) An aggregation of cases of a disease or other health conditions that are closely grouped in time and place. ______
2) The number or proportion of cases or events or conditions in a given population. ______
3) A form of incidence that measures frequency of disease, chronic conditions, or injury in a particular population for a limited time, such as during an outbreak. ______
4) Capable of being transmitted from one person to another by contact or close proximity. ______
5) The amount, or number of occurrences, of a disease, chronic condition, injury, or other attribute or event in a population. ______
6) The period following exposure, when pathologic changes are not apparent and ending with the onset of symptoms of an infectious disease. ______
7) An illness that occurs infrequently and irregularly. ______
8) A group whose members have had contact with a cause of, or possess a characteristic that is a determinant of a particular health problem. ______
9) The occurrence of more cases of a particular type of disease, chronic condition, or injury than expected in a given area, or among a specific group of people over a particular period of time. ______
10) The proportion of people who are exposed to an agent and become infected. ______
11) Consists of qualitative categories whose values have a distinct order. ______
12) Change in physiological status or in the occurrence of a disease, chronic condition, or type of injury that conforms to a regular seasonal pattern. ______
13) A person or animal who harbors the infectious agent for a disease and can transmit it to others, but does not show signs of the disease. ______
14) A factor that must be present for a disease or other health problem to occur. ______
15) The period following exposure, when pathologic changes are not apparent, and ending with the onset of symptoms of a chronic disease. ______
16) The process in which the outbreak is characterized by time, place, and person. ______

Section 2: Multiple Choice *(25 points– 1 point per question)*

1) Which of the following is NOT one of the CDC's top 10 most important public health problems?
   a) HIV
   b) Teen pregnancy
   c) COVID
   d) Heart disease and stroke

2) Which of the following is NOT a type of individual-based analytic study?
   a) Cross-sectional
   b) Health survey
   c) Case-reference
   d) Follow-up study

3) Who is the current director of the world health organization (WHO)?
   a) Halfdan Mahler
   b) Margaret Chan
   c) Robert Redfield
   d) Tedros Adhanom Ghebreyesus

4) What is considered the first step in the control of a disease?
   a) Isolation
   b) Diagnosis
   c) Field Investigation
   d) Quarantine

5) What type of study design is most commonly used in foodborne outbreak investigations?
   a) Case-control study
   b) Randomized clinical trial
   c) Ecological study
   d) Cross-sectional study

6) At the University of Georgia student luncheon, 60 people ate chicken salad. Of the individuals who ate the chicken salad, 45 became ill. 40 people did not eat the chicken salad, and 5 of them became ill. What is the attack rate among those who ate the chicken salad?
   a) 25%
   b) 50%
   c) 75%
   d) 90%

7) A physician notices an unusual cluster of severe respiratory cases in her clinic and treats each patient with antivirals. Meanwhile, the county health department investigates the outbreak, conducts contact tracing, and issues prevention guidelines. Which statement BEST explains the distinction in their approaches?

a) The physician is applying the clinical approach by focusing on individual treatment, while the health department is applying the public health approach by focusing on population-level prevention.
b) The physician is applying the public health approach by focusing on the outbreak cluster, while the health department is applying the clinical approach by recommending antivirals.
c) Both are applying the clinical approach because both involve patient care.
d) Both are applying the public health approach because both involve identifying causes.

8) Which of the following most accurately describes why John Snow's cholera investigation is considered a landmark in epidemiology?

a) It was the first demonstration of the germ theory of disease.
b) It used systematic data collection, mapping, and natural experiments to link an environmental source to disease occurrence.
c) It proved that water filtration removes Vibrio cholerae from water.
d) It used randomization and blinding to test treatment effectiveness.

9) A surveillance system has high sensitivity but low predictive value positive (PVP). Which of the following is MOST likely true?

a) Many true cases are being missed, but most identified cases are real.
b) Most true cases are being detected, but many false positives are also reported.
c) The system captures cases quickly, but data cannot be generalized.
d) The system has high accuracy but poor timeliness.

10) Which stage of the natural history of disease is the target for *secondary prevention*?

a) Susceptibility
b) Subclinical disease
c) Clinical disease
d) Recovery

11) What is the *main* purpose of PulseNet?

a) It creates epidemiological graphs
b) It determines the probability of outbreaks in different regions
c) It maps out the genome of Caenorhabditis elegans
d) It detects outbreaks caused by foodborne illnesses

12) Which of the following is the only criterion considered **absolutely necessary** in Bradford Hill's framework?

a) Specificity
b) Temporality
c) Biological Gradient
d) Analogy

13) Which framework below modified Koch's postulates to account for asymptomatic infections and non-culturable pathogens?

a) Bradford Hill criteria
b) Evan's postulates
c) DAGs
d) GRADE

14) Which of the following best defines **confounding**? [*]

a) A random error in data collection
b) A systematic error that always underestimates disease risk
c) A third variable related to both the exposure and outcome that distorts the association
d) A mistake in laboratory testing

15) Which of the following is an example of **selection bias**?

a) Including controls from a hospital population that differs from the source population of cases
b) Interviewers probing cases more thoroughly than controls
c) Random misclassification of exposure status
d) Laboratory test contamination

16) In an epidemic curve (epi curve), which of the following patterns best indicates a *propagated outbreak*?

a. A single sharp peak followed by a steady decline
b. A flat plateau lasting several incubation periods
c. A series of progressively taller peaks spaced by incubation periods
d. A random scattering of cases with no clear peak

17) Which measure would a hospital administrator MOST likely use to estimate the number of beds needed during an influenza outbreak?

a. Incidence rate
b. Point prevalence
c. Relative risk
d. Attack rate

18) At a community dinner, 200 attendees ate Caesar salad. Of these, 80 people became sick. Among 100 people who did not eat the salad, 10 became sick. What is the relative risk of illness from eating the salad?

a. 0.8
b. 2.0
c. 4.0
d. 8.0

19) Which of the following is a secondary prevention strategy?

a. Childhood immunizations
b. Quarantining exposed travelers
c. Mammogram screening for early breast cancer
d. Physical therapy for stroke patients

20) A disease is always present in a population, but at a relatively stable level. Which term best describes this?

a. Sporadic
b. Endemic
c. Epidemic
d. Cluster

21) Which control measure targets the mode of transmission in the chain of infection?

a. Vaccination of schoolchildren against measles
b. Boiling contaminated water before drinking
c. Screening blood donors for hepatitis C
d. Prescribing antivirals to infected patients

22) A public health team measures the total number of existing diabetes cases in a country at one point in time. Which measure of disease frequency are they calculating?

a. Incidence proportion
b. Cumulative incidence
c. Prevalence
d. Mortality rate

23) Which of the following BEST illustrates the primordial level of prevention?

a. Teaching elementary school children healthy eating habits
b. Administering flu vaccines in a clinic, screening workers in a factory for lead exposure
c. Prescribing statins after a heart attack
d. Screening workers in a factory for lead exposure

24) What is the key weakness of using a sentinel surveillance system?

a. It is too costly and resource-intensive to maintain
b. It relies on an incomplete and variable population since it uses selective groups
c. It may not represent the entire population since it uses selected groups
d. It only detects outbreaks after they are already widespread

25) After implementing a new mosquito-control program, the incidence rate of West Nile Virus in a region dropped from 40 per 100,000 to 10 per 100,000. Which conclusion is MOST appropriate?

a. The strategy reduced prevalence but not incidence
b. The program was ineffective; the decline is random
c. The control program likely reduced transmission
d. The relative risk increased fourfold

<u>Section 3: Free Response</u>

1) What prevention level (primary, secondary, tertiary) is vaccination, and why? *(1 point)*

2) In 1-2 sentences: How would the clinical and public health approaches differ in addressing an opioid overdose crisis? *(2 points)*

3) List the 4 steps in solving health problems in order: *(1 point)*

4) During what stage of subclinical disease is the disease said to be asymptomatic (in terms of infectious diseases, not chronic diseases)? *(1 point)*

5) Name three different possible modes of transmission: *(1 point)*

6) A cohort study reports a *Relative Risk (RR)* **of 2.5** for exposure to disease, with a **95%** *Confidence Interval* of **1.4-4.2.**

   a. Is this result statistically significant? *(1 point)*

   b. How should you interpret the confidence interval? *(1 point)*

7)

a. A hypothetical disease has a basic reproductive number (R0) of 5.

   i. Calculate the minimum percentage of the population that must be immune to achieve herd immunity. *(2 points)*

   ii. If the current vaccination rate is 70% and a small portion of the population (5%) has natural immunity, is the population protected by herd immunity? Why or why not? *(2 points)*

8) Describe the difference between herd immunity for a community and individual immunity. *(2 points)*

9) Why is **specificity** considered the weakest Bradford Hill criterion in modern epidemiology? *(3 points)*
[*]

10) Investigators that are studying smoking and lung cancer find that coffee drinkers appear to have higher lung cancer rates. Explain how **confounding** could explain this finding. *(2 points)*

Section 4: Case Studies

STUDY 1 – Flu on Fly:

On the morning of October 3rd, you and your soccer teammates were flying from Atlanta to Chicago for a big tournament on October 5th. Unbeknownst to you and your fellow passengers, someone on board had the flu, caused by an influenza virus… On the morning of October 5th, 6, including yourself, of the 11 people on your team woke up with sore throats, runny noses, and coughs. Because over half your team is sick, your team loses both the game and the chance to compete at the NCAA, your biggest dream snatched and lost. Since you are the ultimate epidemiologist and revenge seeker, you formulate a plan to track down patient zero (please don't do this in real life).

1. Write a case definition for this outbreak. *(2 points)*

2. Describe any 2 steps of outbreak investigation in detail and how they would pertain to this scenario. *(4 points)*

3. Through various connections, you manage to interview all passengers and flight crew on that flight. You want to separate them into exposed and unexposed groups to determine if exposure increased the chance of contracting the flu. You follow up with all subjects for the next 10 days to see if they contract the flu. What kind of study is this? *(1 point)*
    a) Ecological
    b) Cohort
    c) Cross-sectional
    d) Randomized control

4. Identify 2 benefits and 2 drawbacks to the study type answered in question 2.*(4 points)*

There were some smart people on the plane who wore masks (if only your team had thought of that). Look at the graph below and answer the following questions.

Exposure Status	Contracted Flu	Did Not Contract Flu	Total
Exposed (no mask)	345	2	347
Not Exposed (mask)	3	222	225

5. Calculate the attack rate for both the exposed and not exposed groups on the flight. Then calculate the attack rate for the whole population on the flight. Round answers to the nearest tenth. Be sure to show all work. *(2 points)* [*]

6. Calculate the relative risk and round to the nearest whole number. What does this mean for the exposed group? *(1 point)*

7. You begin to survey just those with confirmed cases and find that only one person displayed classic flu symptoms before boarding the flight. What kind of study design is this? *(1 point)*

8. After finding the person who gave everyone the flu, you berate them for your woes. If you were to teach them how to prevent an outbreak via an airborne agent, what are 2 ways they could do so? Briefly explain how they work. *(2 points)*

STUDY 2 – Summer Camp:

In mid-July, a summer camp reported that 26 of its 120 campers developed acute gastrointestinal illness (vomiting, diarrhea, abdominal cramps) within 24–48 hours of eating a special outdoor picnic dinner. Camp health staff treat affected campers with fluids and supportive care. No campers are hospitalized.

The local health department is notified and begins an investigation. They collect stool samples, interview campers about symptoms and food exposures, and examine the food service operation. The investigation reveals that all affected campers ate potato salad prepared with mayonnaise that was left unrefrigerated for several hours. Laboratory results confirm *Salmonella enterica* in both patient specimens and leftover potato salad.

The health department recommends stricter refrigeration practices and food safety training for camp staff.

1. Which of the following best describes the *clinical* approach in this scenario? *(1 point)*
   a) Testing stool samples to identify Salmonella
   b) Providing oral rehydration to sick campers
   c) Recommending refrigeration of perishable foods
   d) Conducting interviews to identify exposures

2. Which of the following best describes the *public health* approach? *(1 point)*
   a) Providing fluids to sick campers
   b) Identifying Salmonella in stool samples
   c) Recommending refrigeration of foods to prevent future outbreaks
   d) Checking campers' vaccination status

3. Which stage of the natural history of disease did campers enter when they were infected but not yet showing symptoms? *(1 point)*
   a) Susceptibility
   b) Subclinical disease
   c) Clinical disease
   d) Recovery

4. Which type of surveillance was most likely involved in detecting this outbreak? *(1 point)*
   a) Passive
   b) Active
   c) Sentinel
   d) Syndromic

5. Calculate the attack rate of illness among campers. Show your work. *(1 point)*

6. If 10 staff members ate the potato salad but none became ill, how might you explain this discrepancy? *(2 points)*

7. What level of prevention does each of the following represent:

a) Treating sick campus with fluids *(1 point)*
b) Refrigerating food to prevent Salmonella growth *(1 point)*
a) Tertiary
b) Primary

STUDY 3 –:

At a wedding dinner, 200 guests ate chicken, 150 ate fish, and 100 ate both. Illness data:

- Chicken only: 40 sick / 100 total
- Fish only: 10 sick / 50 total
- Both: 70 sick / 100 total
- Neither: 5 sick / 50 total

a) Calculate the attack rates for each group. *(4 points)*

b) Which food is more strongly associated with illness? *(1 point)*

c) Revise the hypothesis about the likely cause of the outbreak. *(2 points)*

Case Study 4 - Salmonella at the Country Fair

On August 12th, a country health department noticed an unusual rise in emergency room visits for diarrhea, abdominal cramps, and fever. Many patients reported attending the Country Fair's Petting Zoo Exhibit two days earlier. Investigators interviewed 200 attendees:

- 120 reported direct contact with animals. Of these, 60 later developed symptoms.
- 80 reported no animal contact. Of these, 10 later developed symptoms.

Laboratory testing confirmed *Salmonella enterica* in stool samples from several patients. The outbreak received media attention, prompting control measures: handwashing stations, educational posters, and temporary closure of the exhibit for deep cleaning. Two weeks later, case counts decreased substantially.

1. Which type of epidemiologic curve pattern would likely appear in this outbreak? *(1 point)*
    a. Continuous common source
    b. Point source
    c. Propagated outbreak
    d. Sporadic outbreak

2. Calculate the attack rate among attendees who had animal contact. *(1 point)*
    a. 25%
    b. 50%
    c. 75%
    d. 80%

3. Which of the following is an example of a primary prevention strategy relevant to this case? *(1 point)*
    a. Providing antibiotics to sick patients
    b. Installing handwashing stations at the exhibition
    c. Investigating stool samples in the laboratory
    d. Rehabilitating patients after severe illness

4. Calculate the relative risk (RR) of developing illness for those who had animal contact compared to those who did not. (Show your work. Remember to include the formula, the formula with the correct values entered, and the answer with units.) *(3 points)*

5. Identify two control strategies used in this outbreak, and briefly explain how each breaks the chain of infection. *(2 points)*

6. a. **Identify the level of prevention (primordial, primary, secondary, tertiary) for each of the following *(2 points)* [*]**

i. Installing handwashing stations:___________________________

ii. Diagnosing and treating patients quickly:_________________________

iii. Teaching children healthy habits around animals from a young age :__________________

iv. Providing rehabilitation for severe long-term complications :__________________

b**. After the intervention, case counts dropped significantly. Identify one strength and one weakness of the prevention strategies used in this outbreak.** *(2 points)*